

From individual sensors to a slow gait

M19 • 04 Jun 2028 - 06 Jul 2028 • 75 hours

What you will build this month

Create a quadruped simulation with state estimation from the IMU and stance-leg kinematics, a deterministic planner for a slow gait that swings legs one at a time (crawl), reachable foothold selection and controlled stopping. The robot must stand for at least 60 seconds and execute a short, reproducible sequence of steps. For every attempt, save contacts, slip indicators, support margin, state errors, failure reasons and whether a fall occurred.

The central engineering task is to connect confidence in sensors with permission for the next step. Even accurate inverse kinematics cannot rescue a gait if the controller uses contact from a lifted leg, confused IMU axes or stale body position. Therefore, first verify the standing robot's state, then swing one leg, and finally execute a full cycle. Increased speed and trotting are outside the required implementation.

What to bring from previous modules

Prepare a model of the body and four legs, forward and inverse coordinate transforms, joint limits, verified actuator simulation and telemetry format. Save masses, centres of mass and the initial stance in a separate configuration. Check simulation stability with the chosen integration step and a floating body. Fixing the body is acceptable in a leg-kinematics experiment, but such an experiment does not demonstrate robot balance.

The M18 ML classifier may run as an observer for comparison with classical contact detection. The required gait operates without it. Simulator ground-truth coordinates and contact forces are used for evaluation and diagnostics; do not silently feed them to an estimator claimed to use only IMU and encoders. List every component's inputs explicitly.

Connection to the future robot

Four legs provide a clear intermediate stage between one mechanical leg and walking; the move to six legs comes later, after the baseline testbed succeeds. M19 does not require rebuilding the physical body or starting push experiments on a real robot. Reproduce all described disturbances in software. The result supports stance-force allocation in M20 and staged transfer to hardware in M22.

The same principle applies to the hand: a supporting finger imposes a kinematic constraint only when contact is reliable. Compare a foot on the ground with a finger on an object: what is known about surface motion, and when does the stationary-surface assumption fail? Gait remains this month's primary task; the grasp state machine is studied separately.

A 75-hour work plan

Four blocks with observable outcomes

Hours 1-20: 10 hours of theory and 10 of practice. Study coordinate frames, IMUs, sensor biases, orientation prediction, kinematic contact constraints and time alignment. Prepare a standing-body state estimate, compare it with ground truth and test exclusion of an unreliable leg. Describe observability limitations. First check signs and units in a simple case, then move to a filter with more states.

Hours 21-40: 10 hours of theory and 10 of practice. Study stance and swing phases, allowed transitions, stance-time fraction, starting, pausing and stopping. Create a slow-gait state machine that swings only one leg and prohibits the next swing when support is insufficient. For now, use a prescribed foot trajectory above a flat surface; add terrain after checking transitions.

Hours 41-60: 10 hours of theory and 10 of practice. Study swing trajectories, clearance, reachability, collisions, surface height, foothold selection with a margin and fallback mode. Implement selection from a small finite set of points, checks along the complete trajectory and a log of rejection reasons. Local selection suffices for a short step; a global terrain planner is unnecessary.

Hours 61-75: 5 hours of theory and 10 of practice. Topics are support margin, body levelling, simplified load distribution, small repeatable disturbances and degraded mode. Deliver a test harness, a 60-second stand, a crawl under declared conditions and a verified pause. Total: 35 hours of analysis and 40 hours of your own experiments.

Organising sessions

Five two-hour sessions prepare each of the first three 10-hour practical blocks. Divide the final 5 analysis hours as 2 + 2 + 1. Split a Sunday practical block into preparation, one implementation, measurement and analysis; schedule breaks separately. Hours below indicate workload sequence. Boundary dates may overlap the next module: place sessions in free periods of the original calendar without duplicating hours; rest, holidays and integration blocks remain unchanged.

Before each run, record a hypothesis and restore an identical initial state. Store the scenario in the configuration: slope, noise, packet loss, leg order and body commands. Video reveals a failure; numerical logs help explain it. Change controller gains, simulation step and geometry separately so that the cause of improvement can be identified.

Orientation, velocity and trust in contact

A minimum state model

Let W denote the world frame, B the body frame, r_i the position of foot i relative to the body in metres, and R the rotation from B to W . The state for initial experiments comprises body orientation, velocity and position, an estimated gyroscope bias and contact-quality flags. A gyroscope measures angular velocity in rad/s with bias and noise; an accelerometer measures specific force, not ready-to-use linear acceleration in the world frame. On a stationary body, readings depend on orientation and the chosen gravity convention.

Task 1. Axis and time conventions

Draw body and leg axes, specify the sign of z and quaternion component order. Record a stationary body, a small positive rotation about each axis and one foot moving forward. Compare measurement signs with the expected motion. Save frames.md and three plots; detect sign errors before introducing the filter. Convert timestamps to a common time base and retain each measurement's age separately.

Task 2. Predict orientation

Estimate the initial gyroscope bias from a stationary interval. Subtract it and integrate angular velocity using the actual dt and a chosen library rotation operation. After each step, check quaternion norm and finite values. Compare with ground truth using the relative rotation angle. Quaternion components cannot be subtracted like ordinary coordinates: q and $-q$ represent the same rotation.

Introduce a constant bias and missing packets. Plot orientation error over time and compare uncompensated and compensated versions. Do not claim that an accelerometer always corrects tilt: during dynamic acceleration, its direction is not simply aligned with gravity. M19 requires only a restricted slow-motion regime and an explicit reduction in confidence when this assumption is violated.

Task 3. Kinematic velocity from a support foot

For a stationary, non-slipping foot, v_B in $W = -R(\omega_B \times r_i + J_i \cdot \dot{q}_i)$, where J_i maps joint velocities to foot velocity in B . Check dimensions: $\text{rad/s} \times \text{m}$ and $\text{J} \cdot \text{rad/s}$ give m/s . Standing should yield velocity near zero; when the body moves with the foot fixed, the estimated velocity sign must match ground truth. Combine several supports using a weighted mean only after rejecting unreliable contacts.

Study the structure of a full invariant EKF as a possible extension. Without an external reference, absolute position and yaw generally drift: contact does not create a global reference. State in the report which local tasks the estimate supports and which absolute-accuracy limitations remain.

Contact, slipping and sensor alignment

When the contact constraint fails

A touching foot is not necessarily stationary relative to the world: it may slip, the surface may move and joints may deflect under load. The estimator needs different confidence weights for different legs. Store planner mode, measured contact evidence, confidence and the time of the last reliable update separately. A scheduled stance phase does not confirm actual contact.

Task 4. Velocity residual during slipping

For each presumed support, calculate the foot-velocity residual: $r_v = v_{est} + R(\omega \times r_i + J_i \cdot \dot{q}_i)$. Its norm is expressed in m/s and should be small for a consistent stationary support. First obtain its distribution in a quiet stance, then during known slipping in simulation. Choose a threshold from these diagnostic data and state its applicable conditions; there is no universal number for every robot.

Where possible, compare a leg with an estimate formed without its own contribution, provided enough other supports exist. Otherwise, explicitly state the residual's dependence on the combined estimate. Test slipping of each leg in turn, a brief impact and loss of all reliable contacts. Save plots of residuals, detection delays and each leg's weight. A brief spike must not cause continuous state chattering.

Task 5. Hysteresis and data loss

Implement a classical contact state machine with separate activation and deactivation conditions, a minimum confirmation time and a timeout. Choose durations from sensor rate, noise and permissible response delay. At low confidence, reduce the constraint weight; without fresh measurements, exclude it entirely. The unknown state means insufficient information and therefore cannot be treated as a confirmed swing phase.

Introduce a known encoder time offset relative to the IMU. Compare residuals and velocity error before and after correct alignment. Then delete an IMU packet and introduce a long gap. A short gap may be handled by prediction within a declared horizon; a long gap puts the system into degraded mode. Record actual data age and transition reason, not only the final flag.

Analysis to document

In the same scenario, compare four versions: IMU without bias compensation, IMU with all contacts unchecked, contact-based estimation with rejection, and estimation with a delayed channel. Use one ground truth. Show velocity root-mean-square error, RMS, maximum tilt error and time without a reliable estimate. Rejection discards some information; explain why error may increase over particular intervals.

Gait as a state machine of permitted transitions

Stance and swing phases

During stance, the foot supports the body; during swing, it moves to the next point. The duty factor is stance duration divided by the leg's period. It is usually high in a slow gait with legs swinging one at a time; specify intervals in a schedule with additional pauses. Trotting swings diagonal pairs and requires a different dynamic stability analysis. Study the distinction this month; enabling trot would require new checks.

Task 6. A deterministic state machine

Define IDLE, SHIFT_BODY, LIFT, SWING, TOUCHDOWN, VERIFY_CONTACT and PAUSE states. In IDLE, first check estimate freshness and sufficient support. SHIFT_BODY moves the centre-of-mass projection into an admissible region of the remaining legs. Lift a leg only after confirming the margin. After SWING, confirm contact by measurement, then transfer load gradually; only after this select the next leg.

Define conditions for every transition: trajectory completion, contact, timeout, insufficient margin or stale state estimate. Errors take priority over ordinary transitions. The state machine receives time, state estimate, contact statuses and a velocity command; it outputs leg modes, target points, a body command and a pause reason. Identical inputs must yield the same sequence regardless of rendering frequency.

Task 7. Starting, stopping and a new command

Run the sequence: rest → slow motion → direction change → stop request during swing → restart. Stopping does not mean instantly setting every joint position to zero. Choose a defined policy: complete an admissible swing and confirm contact, or perform a controlled return if reachable. With insufficient information, maintain the declared conservative mode and do not begin a new step.

Check these rules: at most one leg swings in the normal slow gait; the next leg lifts after the previous contact is confirmed; a command change does not reset the phase backwards; identical messages do not restart the timeout. Save transitions.csv with transition conditions. Reprocessing the recording must produce the same state sequence.

Define the response to losing two supports and being unable to complete a step. Holding a pose may be physically impossible in an unstable configuration, so a stop command alone does not guarantee safety. Mark failure in simulation, stop issuing further gait commands and save the final seconds for analysis.

Foot trajectory and foothold selection

Swing geometry

A trajectory specifies foot position in metres and its time derivatives. Start with smooth interpolation and zero velocity at both ends; define swing height with a separate smooth curve. Explain why a position discontinuity demands high velocity, and a velocity discontinuity demands acceleration. Choose height from the prescribed terrain and its estimation error.

Task 8. Clearance and reachability

On a flat surface, swing one leg along a short arc. Check inverse kinematics throughout the trajectory, joint position and velocity limits, and absence of intersections with the body or other legs. Measure minimum clearance relative to the surface below the foot at every intermediate step. Save plots of height, joint velocities and minimum distances. Even when the endpoint is reachable, the intermediate arc may be inadmissible.

Add a low obstacle in simulation and an error in the height map. Derive the height limit from model geometry and available range of motion. Compare swings with sufficient and insufficient clearance, and retain the collision cause. Raising the foot does not always help: it may move the knee beyond its angle limit or bring the leg closer to the body.

Task 9. Choose a foothold with a margin

Generate a finite candidate grid around the nominal foothold. First filter out candidates that are unreachable, too close to an obstacle or on unknown surfaces. For the remaining candidates, evaluate reachability margin, local slope, surface quality and change in support margin. Normalise quantities into dimensionless scores; otherwise, millimetres may accidentally outweigh degrees purely because of numerical scale.

Save candidates and rejection reasons: `outside_workspace`, `collision`, `insufficient_clearance`, `unknown_height`, `poor_support`. Check the entire trajectory to the selected point. If no suitable point exists, reduce step length under predefined rules; if that also fails, maintain support and enter PAUSE. The fallback algorithm must respect the original limits.

Compare planned and actual touchdown points in one coordinate frame, in metres. Increasing error may come from terrain, delay or actuator tracking accuracy. The candidate-selection algorithm can be retained for a six-legged robot, but step admissibility must be recalculated for its support set.

The crawl_estimator_v1 laboratory

Testbed components and data

Use one simulator and one quadruped model. Processing sequence: time checks → IMU and kinematic estimation → contact checks → gait state machine → foothold selection → foot trajectory → actuator controller. An independent observer saves simulator ground truth and failure indicators. Ground truth must not secretly correct the estimate; otherwise, operation depends on data unavailable to a real robot.

Specify masses, inertias, geometry, physics step, controller and sensor rates, initial stance, leg order, noise, timeouts and motion limits in the configuration. At every tick, record time, true and estimated state, contacts, confidence, residual, measurement age, phase, selected point, failure reason and commands. Save events separately so an incorrect transition can be replayed quickly.

Task 10. Standing and a full cycle

First stand for 60 seconds on a level surface without swings. Check for no fall, finite state values, compliance with joint limits and absence of hidden body fixation. Then perform one cycle that swings each leg, followed by a stop. Only afterwards repeat several cycles at a low commanded speed. Define path length and tilt limits for the specific model before running, and record them as criteria for the educational scenario.

Define a fall explicitly: for example, body contact with the floor, or orientation or height outside selected limits. Justify the boundaries for the model; they are not universal physical standards. Fall rate is the number of episodes containing a fall divided by the number of runs. Account separately for distance travelled and stops caused by deteriorating conditions: stopping after one second does not demonstrate route completion.

Task 11. Compare estimators

Compare identical scenarios in three modes: ground-truth state as a diagnostic reference, IMU-only estimation and contact-aided estimation. Present results for each mode separately. Compare orientation and velocity errors, stable-walking duration and pause count. If the gait works only with ground-truth state, contact-based estimation is not ready: identify how estimation error causes an incorrect transition.

For every failure, save a short recording before and after the event, the configuration and a reproduction command. Then repeat the scenario without rendering and compare contacts: control must keep working when graphical-interface load changes.

Balance, disturbances and deteriorating conditions

What support margin means

For slow motion on a horizontal surface, construct the convex polygon of reliable support points and project the centre of mass onto the plane. The margin is the minimum signed distance from the projection to the polygon edges, in metres. A positive value inside is useful for quasi-static planning; it does not fully guarantee dynamic stability. Sloping or uneven surfaces and appreciable accelerations require additional force conditions.

Task 12. Weight transfer and levelling

Calculate the centre of mass from link masses and positions, rather than just the body's geometric centre. Before lifting a leg, recalculate the polygon without it and smoothly move the body into the admissible region. Check the instant before liftoff, mid-swing and the instant after touchdown. If the margin is insufficient, the step does not begin. Deliver a plot of COM, supports and margin, marking state-machine transitions.

Add a small roll and pitch correction through the desired body pose and stance-leg kinematics. Respect limits on height, rate of change and joint angles. Distribute vertical loads using a simple quasi-static model, checking nonnegativity and total weight. A full QP with friction and torques is studied in the next module; for now, identify conditions in which the simplified model is insufficient.

Task 13. A repeatable disturbance

In simulation, apply a force pulse of known force and duration at a declared point on the body. Impulse equals force multiplied by time, in N·s; torque depends on the moment arm. Start with a small fraction of the model's permissible deviation and increase it only within the educational series. Compare identical disturbances with all four supports and during one swing. Save amplitude, phase and application time.

Measure maximum tilt, minimum support margin, time to return to a defined range and number of new steps after the deterioration signal. Count recovery only after remaining within the range for a specified duration, rather than briefly crossing its boundary. Record demands beyond model capability as failures; falls must remain in the series results.

When conditions worsen, reduce commanded speed, prohibit new swings and choose an admissible way to finish the current swing. If the estimate is unreliable or required supports are lost, stop planning motion. Test this transition separately: it must not produce a command jump or an unreachable pose.

Session plan: hours 1-40

Hours 1-20: state estimation

1-2: review transforms between W, B and the foot, quaternions and angular velocity; label units in five log rows. 3-4: study IMU bias, noise, specific force and gravity. 5-6: derive body velocity from the stationary-foot constraint and check a one-dimensional case. 7-8: study contact modes and observability, and sketch yaw drift. 9-10: plan synchronisation and comparisons of estimator variants.

11-20 - first laboratory: 2 hours for axis conventions and ground-truth recordings; 2 for IMU prediction and bias; 2 for kinematic velocity; 2 for contact confidence and slipping; 2 for comparing missing packets, bias and time offsets. Save a plot and explanation after each experiment. Deliver a body-state estimate with confidence and freshness indicators.

If the formula disagrees with ground truth, check the frames of ω and r , the Jacobian definition and agreement between dt and timestamps before tuning the filter. Increasing filter gains does not resolve a coordinate-frame error. First check a simple known motion, then tune noise smoothing.

Hours 21-40: gait state machine

21-22: draw a stance/swing phase diagram and calculate the schedule's duty factor. 23-24: construct the support region with each leg removed. 25-26: describe transition conditions, timeouts and error priorities. 27-28: work through starting, pausing, stopping and command changes. 29-30: record expected events for three short scenarios, including unconfirmed touchdown.

31-40 - second laboratory: 2 hours for the state machine without physics; 2 for synthetic inputs and transition tests; 2 for swinging one leg in simulation; 2 for a four-leg sequence; 2 for stopping and reprocessing a recording. Start with a known plane and a short step. If necessary, postpone turning and backward motion until the next block, while retaining every stopping mode.

At the end of the block, show leg states on one plot and count simultaneous swings. Even one tick with two legs incorrectly swinging may be invisible in video yet cause a fall. Save the working `classical_crawl_baseline` configuration separately from later terrain experiments.

Sessions and checks: hours 41-75

Hours 41-60: footholds

41-42: smooth trajectories and endpoint velocities; 43-44: clearance and height error; 45-46: workspace and collisions; 47-48: normalised foothold criteria; 49-50: rejection, step reduction and reason logging. Practice 51-60: 2 hours each for the trajectory, IK checks at intermediate points, candidate grid, terrain with error and integration with the state machine. Save an explanation for every rejected step.

61-62: study COM, quasi-statics and support margin. 63-64: body balance and disturbance protocol. Hour 65: stopping criteria, result limitations and handover to M20. Practice 66-75: 2 hours for the 60-second stand; 2 for weight transfer; 2 for a disturbance series; 2 for failure checks; 2 for the results summary, video and reproduction from configuration. The total is exactly 75 hours.

Required test scenarios

1. Zero command. Stand for 60 seconds with the body unfixed. Record estimation errors, minimum height and false-contact count. **2. Unconfirmed touchdown.** Remove the surface below the target foot. On timeout, prohibit the next swing even when the trajectory is complete. Check the pause reason and absence of a second swinging leg.

3. One slipping support. Change friction beneath one foot while keeping other conditions fixed. Measure detection delay, number of incorrectly trusted samples and effect on body velocity. **4. IMU/encoder gap.** First introduce a short gap, then one exceeding the freshness limit. Check prediction within the permitted window, then degradation and recovery without a jump.

5. Unreachable foothold. Place all candidates outside the workspace or near an obstacle. Expect rejection reasons, bounded step reduction and PAUSE. **6. Height error.** Change the surface relative to the map. Check clearance and early and late touchdown; the next step must await contact confirmation.

7. Stopping during swing. Request a stop at the start, middle and end. Compare transition admissibility and continuity of target positions. **8. The same push in different phases.** Repeat a known impulse in standing and on three supports. Show maximum tilt, recovery time and fall rate; include both cases in the report.

Acceptance checks, limitations and the transition to M20

What to present as the result

The `crawl_estimator_v1` package includes the model, configurations, estimation code, state machine, foothold checks, test scenarios, source recordings and a reproducible report. Attach video of a 60-second stand and a short gait, with plots. State physics parameters, the floating body and the exact conditions for successful operation, falling or entering a pause.

State estimation: coordinate frames, units, effects of bias, missing data and delays, contact residuals and operation without machine learning have been checked. Orientation error is measured by relative rotation, and velocities are compared in a common frame. Gait: transitions are admissible, the next swing depends on actual support, and starting and stopping have been tested in different phases. Footholds: rejection reasons are saved, and both endpoint and complete trajectory satisfy geometric limits.

Scenario acceptance: stable standing for at least 60 seconds, with episode count, falls, slips, pauses and errors recorded. At least three repeats of each selected nominal and disturbed scenario are useful as an educational minimum; they do not establish real-robot failure probability. If 60 seconds are not achieved, do not declare the module complete after one successful step: save a minimal reproducible case and fix the baseline control loop.

Questions to explain independently

1) Why does touching not imply a stationary support? 2) Why can the commanded phase not be the sole contact ground truth? 3) How does the sign of $\omega \times r$ relate to body motion? 4) Why do yaw and absolute position drift? 5) How does brief packet loss differ from a stale estimate? 6) Why does a positive support margin not guarantee stability during fast motion?

7) How do link centres of mass affect weight transfer? 8) Why does endpoint reachability not guarantee an admissible trajectory? 9) What happens after late touchdown? 10) Why can increasing swing height cause a collision? 11) How does stopping during swing differ from a zero velocity command? 12) How can recordings alone distinguish an estimator failure from a controller failure?

Reduce scope while preserving the objective

If time is short, remove trotting, complex terrain, learned confidence estimation and the full invariant EKF. Retain contact-aided estimation, a flat surface, a limited set of footholds, a slow gait and failure tests. A 60-second stand, actual contact, replay and stopping are mandatory. Hand active supports, COM and orientation estimates, joint limits and a verified classical mode to M20 for force allocation.

Assigned reading: estimation and slow gait

Read the specified extracts before the corresponding tasks. Core reading explains principles; reference reading assists implementation. Sources may be in English; prepare your explanations and report in English. Reading is included in the 75 hours.

Core reading

1. Kevin Lynch, Frank Park, Modern Robotics. [§3.2.1 Rotation Matrices](#); [§3.2.2 Angular Velocities](#); [§5.1.1-5.1.2 Jacobians](#). For Tasks 1-3: reconstruct coordinate-frame transforms and foot velocity from joint motion. The key outcome is to derive the stationary-support constraint including body motion.

2. Simo Särkkä, Bayesian Filtering and Smoothing, 1st ed., 2013. [§4.3 Kalman filter](#); [§5.2 Extended Kalman filter](#). For Tasks 2-5: write the prediction, measurement residual and correction for a small example. EKF is reference material for extension; a full invariant filter is unnecessary.

3. Russ Tedrake, Underactuated Robotics (MIT, course notes). [Planning and Control through Contact: Autonomous Hybrid Systems](#); [Given a fixed mode sequence](#). For Tasks 6-7: describe stance and swing through modes, transition conditions and events. From [Simple Models of Walking and Running](#), read Simple Models of Walking and Stance Dynamics: explain how your slow gait differs from dynamic walking.

Documentation for implementation checks

[SciPy Rotation: from_quat, apply, inv, composition](#) - independent rotation checks in Task 1; explicitly specify quaternion component order and transform direction.

[MuJoCo Computation: Contact, friction, constraint model](#) - only if MuJoCo is chosen: meaning of contact in ground truth and friction parameters. If retaining Gazebo, use documentation for its pinned physics engine.

After reading

Draw the supports, centre of mass and one unreliable foot. Explain which measurements the estimator has lost and why the state machine prohibits the next swing. Copying a gait from someone else's SDK does not replace this reasoning; the required deliverable remains your own small crawl_estimator_v1.

Fundamental concepts in M19

State. The set of model quantities needed to predict motion; here it includes orientation, body velocity and sensor-error parameters.

Orientation. Rotation of the body coordinate frame relative to the world frame; specify the matrix or quaternion together with its transform direction.

Specific force. Acceleration relative to free fall, measured by an accelerometer; in the ideal model $f = R^T(a - g)$, in m/s^2 .

Sensor bias. A systematic measurement-error component that may vary over time; integrating it causes the motion estimate to drift.

Prediction. Propagating a state estimate forward using the dynamic model and inputs, while accounting for model and measurement uncertainty.

Measurement update. Refining a prediction with a new measurement according to its reliability; false contact can corrupt the body-state estimate.

Innovation. The difference between measured and predicted values; it helps detect inconsistent sensors or a model error.

Covariance. A matrix describing error dispersion and pairwise relationships; it characterises uncertainty in the chosen state coordinates and units.

Observability. The ability to distinguish states through measurements and motion; unobservable components cannot be uniquely recovered from those data.

Leg kinematic odometry. Estimating body motion from joint measurements and leg geometry under an explicitly stated assumption about support motion.

Contact. Interaction between a foot and a surface; touching alone confirms neither absence of slipping nor sufficient load-bearing capacity.

Slip. Relative foot motion along a surface; it violates the stationary-support assumption and may distort estimated body motion.

Support polygon. The convex hull of support points in a chosen plane; a geometric guide to quasi-statics when contact assumptions hold.

Static support margin. The minimum distance from the COM projection to the polygon boundary, positive inside; it does not fully account for dynamics and friction.

Stance phase. The part of a leg cycle in which it should support the body; actual contact suitability is checked using measurements.

Swing phase. The part of a cycle that moves the foot to a new support; a swing command does not itself confirm actual absence of contact.

Duty factor. The fraction of the gait period allocated to stance for a leg; it describes time, not force.

Gait phase. A normalised coordinate within the cycle that specifies leg-action order; actual transitions also require event and condition checks.

Transition guard. A logical condition permitting a state-machine mode change, such as reliable contact and an admissible configuration of the other supports.

Foothold. The selected location for the next foot contact, checked for reachability, surface conditions, trajectory clearance and stability margin.

Equipment and software for M19

Use existing equipment

MacBook Pro M4, the four-leg model, simulator contacts, actuator parameters and the Python environment from previous months. Keep the existing physical IMU, MCU and assembled leg; simulation suffices here for disturbances and slow gait. The main ROS route is Ubuntu 24.04 ARM64 in UTM, Jazzy and Gazebo Harmonic.

No mandatory purchases

Do not order a quadruped body kit, additional actuators, a new IMU or an onboard computer. The 60-second stand and step sequence are tested in simulation; purchases for the physical quadruped stage are deferred to M22.

Install now: only what is missing

No new software is mandatory: reuse pinned NumPy, SciPy, Matplotlib, pytest and the current simulator. For transforms, use the installed [SciPy Rotation](#) or your own verified implementation. Maintain one timestamp source and one coordinate-frame description format throughout the testbed.

Optional: a separate MuJoCo route

Only if MuJoCo has already been selected for this laboratory, add mujoco in a separate macOS ARM environment following the [official Python bindings instructions](#). Check build availability for your Python version and ARM64; do not replace a working Gazebo without a reason. On macOS, passive viewing requires launching through mjpgpython.

Before using contact ground truth, read [Computation: contact and constraints](#). Begin model checks without a graphical interface, then enable viewing. You may read CHAMP and other SDKs to compare architectures; they are not mandatory dependencies for your own FSM.

Installation checks and model limits

Restart the scene and check the floating body, masses and gravity. Perform a 90° rotation, inverse transformation and quaternion-order check. Record 60 seconds of standing, then remove one contact. The log must show the change in confidence and whether swing is permitted or prohibited.

Repeat with a stale timestamp, gyroscope bias and foot slipping. On the CPU, measure loop rate and deadline misses separately from UTM graphics load. Simulator ground-truth position and orientation are used only for quality evaluation; the estimator receives the declared sensor data.

The next purchasing decision

A GPU and Jetson are unnecessary for slow gait. The separate G3 on 27.11-10.12.2028 remains a future decision; having ML from M18 does not replace it. Preparation and checks use part of the original 75 hours; equipment-delivery time is excluded.

Motion fundamentals: weight transfer

Required analysis: why admissible support changes before a leg is lifted

What supports the body during slow gait

First check the whole robot's centre of mass: $\text{COM } c = \sum m_j c_j / \sum m_j$. It may differ from the body's geometric centre. All positions below are in world frame W : x forward, y left and z up; units are metres and kilograms. In quasi-statics, neglect accelerations and external loads other than gravity. Feet remain stationary on one horizontal plane; ground reaction forces act on the robot.

Let $F_i \geq 0$ denote the vertical support reaction acting on a foot in N , m the total mass and $g = 9.81 \text{ m/s}^2$. Force balance gives $\sum F_i = mg$; moment balance gives $\sum F_i \cdot p_i \cdot xy = mg \cdot c_{xy}$. Thus $c_{xy} = \sum \lambda_i p_i \cdot xy$, where $\lambda_i = F_i / (mg) \geq 0$ and $\sum \lambda_i = 1$. This explains the convex support polygon: a negative reaction would require an ordinary foot to pull on the ground.

Three supports: calculate before lifting the fourth

Use an example robot mass of 10 kg; lift foot $D = (0.20, 0.15) \text{ m}$. The remaining feet are $A = (-0.20, -0.15)$, $B = (0.20, -0.15)$, $C = (-0.20, 0.15) \text{ m}$. After `SHIFT_BODY`, let $c_{xy} = (-0.05, -0.03) \text{ m}$. From the coordinates: $\lambda_B = 0.15/0.40 = 0.375$, $\lambda_C = 0.12/0.30 = 0.400$ and $\lambda_A = 0.225$. Loads **$F_A = 22.07$, $F_B = 36.79$, $F_C = 39.24 \text{ N}$** sum to 98.10 N.

The diagonal boundary BC is $3x + 4y = 0$; the expression is negative inside the triangle. Distances to the three boundaries are 0.15, 0.12 and $|3(-0.05) + 4(-0.03)|/5 = 0.054 \text{ m}$. **The margin is 5.4 cm.** With the previous $c_{xy} = (0.05, 0.03)$, $\lambda_A = -0.225$: this three-support stance requires a negative load, even though the projection may still lie inside the four-support rectangle.

What to check in the state machine, beyond the diagram

A. In Task 12, calculate c from the links before lifting each leg and recalculate the region without that leg. Save `support_case.csv`: points, c , λ , loads and signed margin. Checks: $\sum \lambda = 1$, reconstruction of both coordinates of c , $F_i \geq 0$ and weight balance; the negative case must prohibit LIFT. Attach the diagram to the existing transitions plot.

B. In Task 6, use an example gait period $T = 4 \text{ s}$ and stance time of 3.2 s: the duty factor is 0.8. Schedule four swings of 0.8 s without overlap, leaving pauses on four supports. The timer specifies the plan; actual contact, a fresh estimate and support margin authorise transition to the next leg. Test unconfirmed touchdown: the schedule must not start the next step.

A positive margin checks geometry under the stated assumptions. Sufficient friction and actuator torques, a load-bearing surface and leg reachability are also needed. Under acceleration, $\sum f + m \cdot g_W = m \cdot cdd$, where $g_W = (0, 0, -9.81) \text{ m/s}^2$; moments change angular momentum. The same polygon therefore does not guarantee balance under a push, during running or slipping, or with non-coplanar contacts.

Time: 1 hour from 23-24 on excluding a leg, and 1 hour from 66-75 allocated to weight transfer. This calculation supports Tasks 6 and 12 within the existing laboratory; all previous tests and 75 hours remain. Read: Tedrake, [Highly-articulated Legged Robots](#), “The special case of flat terrain”; [Planning and Control through Contact](#), “(Autonomous) Hybrid Systems”.